

For official use only:

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Marks: 60	Date: 23 rd May, 2025
Instructions	
- Attempt all questions - Use only the space provided here to answer, so write carefully and legibly. - Understanding the question paper is part of the exam, so no explanations will be offered on questions.	

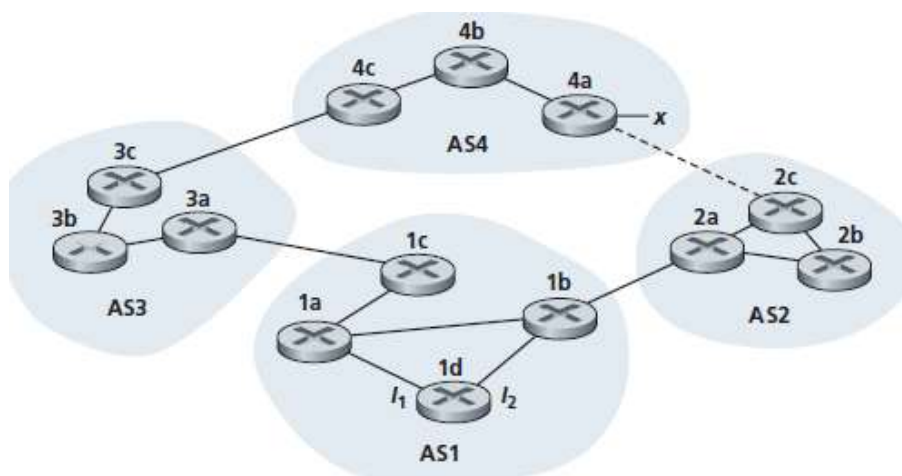
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PART B: Detailed Question/Answers Section (Attempt all questions)**1. Control Plane**

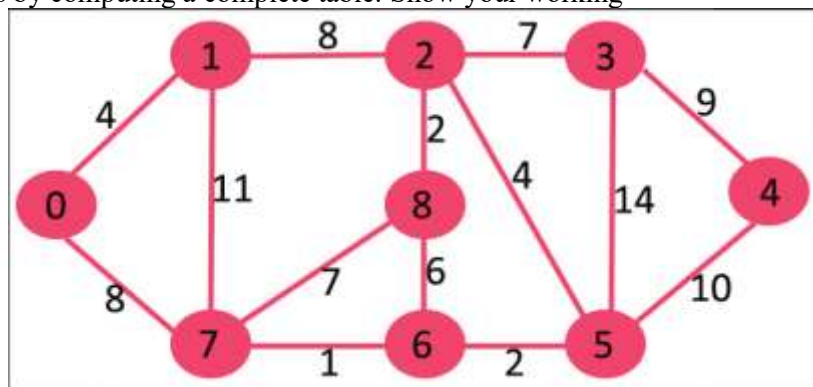
1.1. Consider the network shown below. Suppose AS3 and AS2 are running OSPF for their intra-AS routing protocol. Suppose AS1 and AS4 are running RIP for their intra-AS routing protocol. Suppose eBGP and iBGP are used for the inter-AS routing protocol. Initially suppose there is *no* physical link between AS2 and AS4. (4)

- Router 3c learns about prefix x from which routing protocol? →
- Router 3a learns about x from which routing protocol? →
- Router 1c learns about x from which routing protocol? →
- Router 1d learns about x from which routing protocol? →



- 1.2. What types of messages flow across an SDN controller's northbound and southbound APIs? Who is the recipient of these messages sent from the controller across the southbound interface, and who sends messages to the controller across the northbound interface? (3)

- 1.3. Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from 7 to all network nodes. Show how the algorithm works by computing a complete table. Show your working (5)



2. Data Plane

- 2.1. Consider a datagram network using 32-bit host addresses. Suppose a router has four links, numbered 0 through 3, and packets are to be forwarded to the link interfaces as follows: (4)

Destination Address Range	Link Interface
11100000 00000000 00000000 00000000 through 11100000 00111111 11111111 11111111	0
11100000 01000000 00000000 00000000 through 11100000 01000000 11111111 11111111	1
11100000 01000001 00000000 00000000 through 11100001 01111111 11111111 11111111	2
otherwise	3

- a. Provide a forwarding table that has five entries, uses longest prefix matching, and forwards packets to the correct link interfaces.

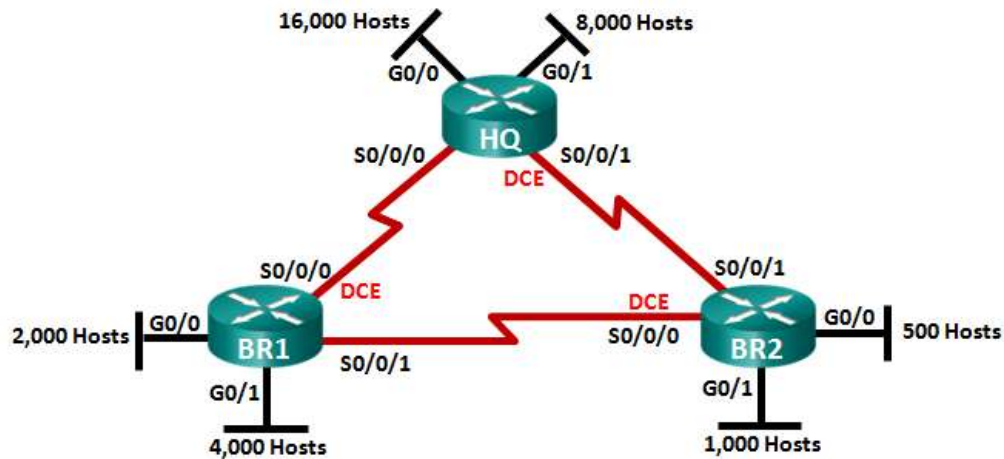
- b. Describe how your forwarding table determines the appropriate link interface for datagrams with following destination addresses:

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11001000 10010001 01010001 01010101
11100001 01000000 11000011 00111100
11100001 10000000 00010001 01110111

```

2.2. Given the following topology examine the network requirements and develop the VLSM address scheme using 172.16.128.0/17 network address. Populate the given tables (re-write them as it is in your answer booklet). Bear that we would adhere to the principle that first host address of the network will be assigned to the router. Likewise, HQ interfaces would have priority over others and BR1 interface over BR2 for assigning IP addresses to interfaces over serial links. Show your complete working or else your answer will be annulled (8)



Subnet Tables:

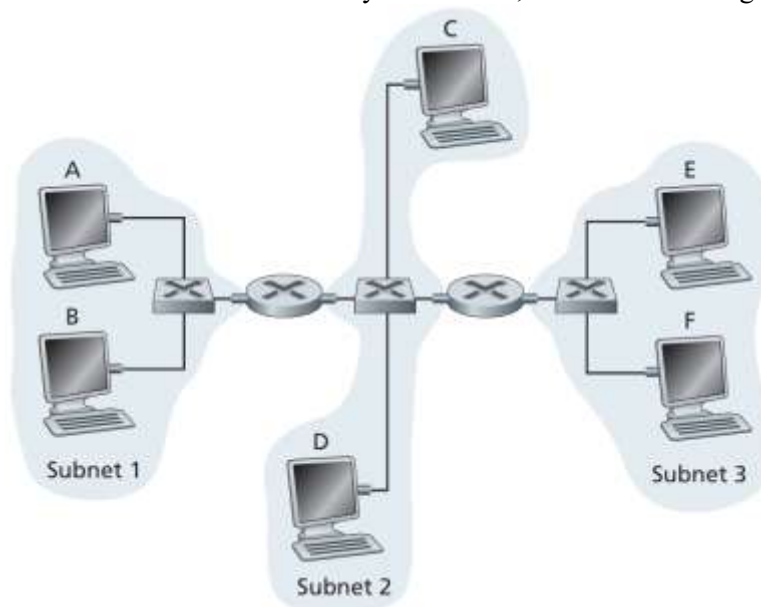
Subnet Description	Number of Hosts Needed	Network Address /CIDR	First Host Address	Broadcast Address
HQ G0/0	16,000			
HQ G0/1	8,000			
BR1 G0/1	4,000			
BR1 G0/0	2,000			
BR2 G0/1	1,000			
BR2 G0/0	500			
HQ S0/0/0 – BR1 S0/0/0	2			
HQ S0/0/1 – BR2 S0/0/1	2			
BR1 S0/0/1 – BR2 S0/0/0	2			

Device	Interface	IP Address	Subnet Mask	Device Interface
HQ	G0/0			16,000 Host LAN
	G0/1			8,000 Host LAN
	S0/0/0			BR1 S0/0/0
	S0/0/1			BR2 S0/0/1
BR1	G0/0			2,000 Host LAN
	G0/1			4,000 Host LAN
	S0/0/0			HQ S0/0/0
	S0/0/1			BR2 S0/0/0
BR2	G0/0			500 Host LAN
	G0/1			1,000 Host LAN
	S0/0/0			BR1 S0/0/1
	S0/0/1			HQ S0/0/1

3. The Link Layer and LANS:

3.1. Consider three LANs interconnected by two routers, as shown in the figure below:

(4)



- Assign IP addresses to all of the interfaces. For Subnet 1 use addresses of the form 192.168.1.xxx; for Subnet 2 use addresses of the form 192.168.2.xxx; and for Subnet 3 use addresses of the form 192.168.3.xxx. **[label the addresses in the figure, clearly, using arrows]**
- Assign MAC addresses to all of the adapters. **[again label them below IP addresses where required]**

- c. Consider sending an IP datagram from Host E to Host B. Suppose all of the ARP tables are up to date. Enumerate all the steps.

- d. Repeat (c), now assuming that the ARP table in the sending host is empty (and the other tables are up to date).

- 3.2. Consider the 5-bit generator, $G=10011$, and suppose that D has the value 1010101010. What is the value of R? Show your working. (4)

4. Transport Layer:

- 4.1. In the generic SR protocol, the sender transmits a message as soon as it is available (if it is in the window) without waiting for an acknowledgment. Suppose now that we want an SR protocol that sends messages two at a time. That is, the sender will send a pair of messages and will send the next pair of messages only when it knows that both messages in the first pair have been received correctly. Suppose that the channel may lose messages but will not corrupt or reorder messages.

Design an error-control protocol for the unidirectional reliable transfer of messages. Give an FSM description of the sender and receiver. Describe the format of the packets sent between sender and receiver, and vice versa. If you use any procedure calls other than `udt_send()`, `start_timer()`, `rdt_rcv()` etc. that we covered in the text, clearly state their actions. Give an example (a timeline trace of sender and receiver) showing how your protocol recovers from a lost packet. **(5)**

- 4.2. Host A and B are directly connected with a 100 Mbps link. There is one TCP connection between the two hosts, and Host A is sending to Host B an enormous file over this connection. Host A can send its application data into its TCP socket at a rate as high as 120 Mbps but Host B can read out of its TCP receive buffer at a maximum rate of 50 Mbps. Describe the effect of TCP flow control. (3)

5. Application Layer:

- 5.1. Consider sending over HTTP/2 a Web page that consists of one video clip, and five images. Suppose that the video clip is transported as 2000 frames, and each image has three frames. (5)
- a. If all the video frames are sent first without interleaving, how many “frame times” are needed until all five images are sent?

- b. If frames are interleaved, how many frame times are needed until all five images are sent.

- 5.2. Why do HTTP, SMTP, and IMAP run on top of TCP rather than on UDP? Illustrate by making a matrix of features required by these applications. (3)

6. Life-long learning:

- 6.1. You are tasked with conducting an OSINT investigation on a fictional individual suspected of running a covert online operation. Using only legal and ethical OSINT techniques, outline your step-by-step approach to gather relevant information. Specify the tools or platforms you would use, the types of data you would look for, and how you would verify the accuracy of the information collected. Finally, discuss the ethical boundaries you must consider during this investigation. (6)

Name: _____

ERP: _____

6.2. How did engaging with standards organizations (e.g., joining ISOC) and learning about lengthy process of standard evolution through their timelines change your understanding of their role? (3)

6.3. What challenges might arise without these organizations? Enumerate. (3)